EXPLORING EFFECTIVE DATA AUGMENTATION WITH TDNN-LSTM NEURAL NETWORK EMBEDDING FOR SPEAKER RECOGNITION

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ABSTRACT

The speaker characterization using four different data augmentation methods and time delay neural networks and long short-term memory neural networks (TDNN-LSTM) is proposed in this paper. The proposed data augmentation is used to increase the amount and diversity of the training data including adding speed perturbation, adding volume perturbation, adding room impulse responses, and adding additive noises. The idea of TDNN-LSTM based speaker embedding is better to capture the temporal information in speaker speech than the conventional TDNN based x-vectors. The proposed methods were trained on VoxCeleb dataset and tested with Speakers In The Wild (SITW) dataset in the evaluation core-core condition. We achieved results of EER=1.86% and a minimum decision cost function (DCF) of 0.204 at p-target=0.01, and a minimum DCF of 0.368 at p-target=0.001. The proposed methods outperform the baselines of both i-vector and x-vector.

Index Terms— Data augmentation, speaker recognition, speaker embedding, TDNN-LSTM, neural network

1. INTRODUCTION

Speaker recognition can be classified into two categories including text-dependent and text-independent tasks. The idea of speaker embedding is to find representation for speaker idiosyncrasy, which can be extracted for both enrollment data and test utterance, and then used to make a decision regarding true speaker or an imposter in speaker recognition. The embedding-based speaker recognition systems often demonstrate sound performance and become the mainstream methods.

In text-dependent speaker recognition, Variani et al. used a fully connected deep neural network (DNN) to extract a speaker discriminative feature, d-vector, from each utterance [1]. Utterance d-vectors are computed by averaging the activation vectors of the last hidden layer for all frames of an utterance. Chen et al. proposed a modified d-vector by replacing a fully connected deep neural network with locally connected DNNs and convolutional neural networks (CNN), which result in shrunk model size for a better performance [2]. Heigold et al. proposed end-to-end speaker recognition by using an LSTM with a single output to represent speaker [3].

The common text-independent speaker verification systems include i-vector [4]-[5] and DNN-based representation [6]. The approach of i-vector dominates systems of speaker recognition from 2010 to 2016. A speaker embedding method based on deep neural network (DNN) has been proposed in 2016 [7]. The proposed architecture was a feed-forward DNN and it outperformed the conventional i-vector. Snyder et al. investigated replacing i-vector for text-independent speaker verification with embedding extracted from a feed-forward deep neural network [8]. In addition to feed-forward neural networks, the convolutional neural networks based speaker recognition was proposed in 2017 [9]. In 2018, data augmentation was used to improve the performance of DNN embedding for speaker recognition, and variable-length utterances are converted to fixed-dimensional embedding vectors, called x-vector [10], which were trained to discriminate between speakers. The x-vector is based on the time-delayed neural network (TDNN) structure [11].

In this paper, we proposed a new speaker embedding neural network architecture which is based on TDNN and long short-term memory (LSTM) recurrent neural networks (RNN). The motivation of using both TDNN and LSTM is to better capture the temporal information in speech than using TDNN alone as in x-vector. In theory, the neural network speaker embedding approaches, such as x-vector and d-vector, can outperform the i-vector approach only when trained on large amounts of data [12]. As a result, four different data augmentation methods are used to increase the amount and diversity of the available training data in this paper. We investigated methods including adding room impulse responses, adding speed perturbation, adding volume perturbation, and adding additive noises. We explored step-by-step improvements by using different data augmentation methods. To the best of our knowledge, we are the first to verify the success of speed and volume perturbation in speaker recognition. Also, we presented the best result on the SITW evaluation set in the core-core condition.

The rest of this paper is organized as follows. Section 2 introduces the proposed methods and speaker characterization framework. Section 3 presents the experimental results and analyzes the details. Conclusions are finally drawn in Section 4.

2. EFFECTIVE DATA AUGMENTATION WITH TDNN-LSTM SPEAKER EMBEDDING

Because the neural network based speaker embedding methods are always data greedy, four different data augmentation methods are further investigated to build robust speaker recognition systems. The proposed data augmentation is applied to create 7 copies of original data (feature vectors) to increase the amount and the diversity of available data. In addition, a TDNN-LSTM speaker embedding is explored to capture better temporal information for speaker recognition in this study.

2.1. Effective data augmentation

The VoxCeleb[1] dataset is the combination of VoxCeleb 1 [9] and VoxCeleb 2 [16] in the training list of TDNN-LSTM. In addition, the training VoxCeleb dataset is used for centering, LDA, probabilistic linear discriminative analysis (PLDA), and speaker embedding neural networks training. The SITW evaluation set in the core-core condition is used to evaluate results in this study. There are 60 overlap speakers between VoxCeleb and SITW which need to be removed from the training list. The overall VoxCeleb dataset contains over 2,000 hours, over 1 million speech utterances for over 7,000 celebrities, extracted from videos uploaded to YouTube. All the audio samples are 16 kHz and in 16-bit format wideband microphone speech for feature extraction. The average number of utterances per speaker is about 170. However, most of these utterances are very short.

Data augmentation is usually used to increase the amount and diversity of the available training data [17]. Four different data augmentation methods are explored in this study to create 7 copies of original data (feature vectors), including adding speed perturbation, adding volume perturbation, adding room impulse responses, and adding additive noises.

2.1.1. Adding speed perturbation

The speed perturbation method is applied to create two more copies of the original signal with speed factors of 0.9 and 1.1 [18]. The SoX[2] speed function is used to modify the speed to 90% and 110% of the original speaking rate. Because the speed perturbation results in a change in the duration of the original signal which affects the number of frames in the front-end feature, the VAD needs to re-estimate based on the modified signal. We can keep using the same VAD information of the original signal for other data augmentation methods which do not affect the duration of the signal.

2.1.2. Adding volume perturbation

Performing volume perturbation of the original training data, where each recording in the training data is scaled with a random variable drawn from a uniform distribution over a range, emulates mean shifts in the Mel-frequency cepstral coefficient (MFCC) domain [11]. The scaling factor for each utterance is randomly chosen from a range (e.g. [0.5, 1.5]). We adopt the volume perturbation by using the sampled scaling factors and increasing the number of available training data for speaker recognition. The improvement is reported on the later experiments. One copy of the original data is created by adding volume perturbation. Both speed perturbation and volume perturbation are common in training automatic speech recognition but not common in speaker recognition. Since neural network based speaker embedding is data greedy, we explore the effects of adding speed and volume perturbation for the robust speaker recognition.

2.1.3. Adding room impulse responses

The simulated room impulse responses (RIRs) [10] are applied to corrupt the original audio by convolving with simulated RIRs. The RIRs is a database of simulated and real room impulse responses, isotropic and point-source noises, which is online available at OpenSLR[3]. Sounds of the size of the small and medium room are included and used for the simulated room impulse responses. The ranges from which the width and length of a room are uniformly sampled are 1m-10m and 10m-30m, respectively. One copy of original data (feature vectors) is created by using the room impulse responses.

2.1.4. Adding additive noises

Besides adding room impulse responses, we can also add additive noises. The MUSAN[4] dataset [19] is used to corrupt the original audio files with additive noises including babble noise, general noise, and music noise, resulting in a 3-fold data augmentation. The MUSAN dataset is an online available corpus of music, speech, and noise which is online download available.

The proposed TDNN-LSTM neural network speaker embedding is trained on the VoxCeleb dataset and evaluated on the Speakers in the Wild (SITW) dataset [14]-[15]. Recordings from the SITW dataset consist of video audio from 299 native English speakers with naturally occurring noises, reverberation, and device variability. The amount of enrollment and test speech is expected to range from 6 seconds to 240 seconds in the SITW dataset. The speaking

[1] http://www.robots.ox.ac.uk/~vgg/data/voxceleb
[2] http://sox.sourceforge.net
[3] http://www.openslr.org/28
[4] http://www.openslr.org/17

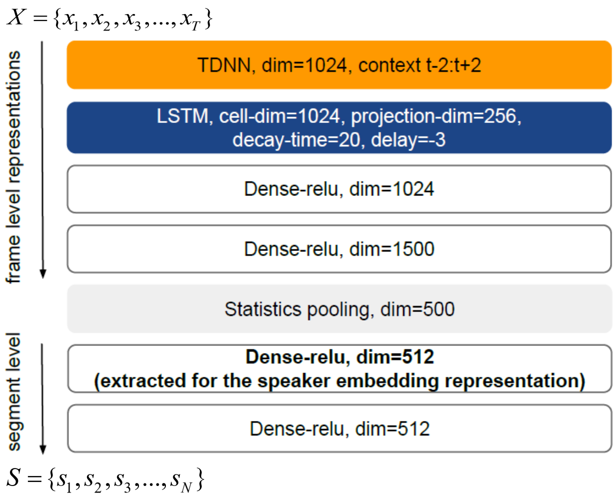

Fig. 1. The proposed TDNN-LSTM neural network structure for extracting speaker embedding representations.

conditions include monologues, interviews and more conversational dialogues with dominant backchannel and speaker overlap [14]. There are totally 721,788 trials on the SITW evaluation set in the core-core condition. There is no data augmentation for both enrollment and testing of the SITW dataset.

Augmenting the original audio with the corrupted copies creates more than 8 million utterances for the VoxCeleb dataset. In total, there are about 9 million utterances available for training after the proposed data augmentation methods. To process such a large amount of data, we throw away the speakers with fewer than 8 utterances and remove features that are too short after removing silence frames. We require at least 400 frames per utterance for training.

2.2. Front-end feature extraction

The MFCC feature is extracted from audio files. MFCCs are computed using 30 Mel filter banks. The audio samples are coded with a 25-ms frame window and a 10-ms frame shift. In addition, pitch features show good discriminative information to recognition speaker characteristics [13]. With the additional three-dimensional pitch features, we use 33 dimensions of MFCC and pitch features for the front-end feature extraction. After doing feature extraction, energy-based voice activity detection (VAD) [20] is used to estimate frame-by-frame speech activity, and the frames with silence or low signal-to-noise ratio in the audio samples are removed for speaker recognition. VAD is important to speaker recognition. In speech recognition [21], there is no need to remove the silence frames, because silence or filter models are trained to detect the non-speech segments. In the audio, the speech decoder will identify them as a short pause or a stop. Such information is helpful for the later natural language understanding and makes speech-to-text results with the sentence structure. However, in speaker recognition, the frames with silence or low signal-to-noise ratio belong to noisy features which hurt the performance of speaker characterization.

2.3. TDNN-LSTM speaker embedding training

We proposed a TDNN-LSTM neural network based speaker embedding for speaker recognition layer as shown in Fig. 1. For training and test speaker-embedding representation extraction, a TDNN-LSTM neural network of 6 hidden layers with rectified linear unit (relu) non-linearity is trained to discriminate thousands of speakers in the training set with millions of utterances. The input of networks is a sequence of speech feature $X = \{x_1, x_2, x_3, ..., x_T\}$ with a size of *T*. The output of networks is a vector of speaker label $S = \{s_1, s_2, s_3, ..., s_N\}$ with a size of *N*. The neural networks have 6 hidden layers (the structure of TDNN-LSTM-Dense-Dense-Dense-Dense), an input dimension of 33, and an output dimension of 22,089. The first hidden layer is a 1,024 dimensional TDNN with the context information of t-2 and t+2. The second hidden layer is a LSTM with cell-dimension=1,024, projection-dimension=256, decay-time=20, and delay=-3. Beside TDNN and LSTM, Dense-relu layers are shown from the third layer to the 6th layer. The first 4 hidden layers operate at frame-level, while the last 2 layers operate at segment-level. There is a statistics pooling layer between the frame-level and the segment-level layers that aggregates all frame-level outputs from the 4th layer and computes the mean and standard deviation over all frames for an input segment. The long-term speaker characteristics can be captured in the network by a temporal pooling layer that aggregates over the input speech. After the neural network training, speaker-embedding representations are extracted from the 512 dimensional affine components of the 5th layer, which is the first segment-level layers.

2.4. Back-end processing

The extracted TDNN-LSTM speaker embedding representation is centered by the mean and projected to 150 dimensionalities using LDA. In addition, the length normalization and PLDA are applied to show discriminative characteristics. The centering mean, LDA and PLDA are trained using the VoxCeleb dataset with data augmentation.

3. EXPERIMENTS

The proposed methods are evaluated with performance metrics of the equal error rate in the percentage (EER%) and the minimum of the detection cost function (DCF) with the different prior target probability (p-target). The p-target ratio of 0.01 is defined as DCF1. The p-target ratio of 0.001 is

Table 1. Step-by-step improvements by using different methods on SITW evaluation set in the core-core condition. Our contributions are data argumentation and TDNN-LSTM neural network embedding. The best result was boldface.

Methods	Network type	# of utterances	EER (%)	DCF1	DCF2
i-vector baseline [4]	-	6,155,850	5.659	0.463	0.629
x-vector baseline [10]	**TDNN**	**6,155,850**	**3.499**	**0.342**	**0.516**
original speech only	TDNN	1,231,170	4.456	0.428	0.627
+reverb	TDNN	2,462,340	4.183	0.395	0.591
+speed	TDNN	4,924,680	3.964	0.322	0.478
+volume	TDNN	6,155,850	3.827	0.312	0.477
+noise	**TDNN**	**9,849,360**	**3.253**	**0.289**	**0.443**
The proposed methods	**TDNN-LSTM**	**9,849,360**	**1.859**	**0.204**	**0.368**

noted as DCF2 in this study. The results are implemented using the open-source Kaldi speech recognition toolkit [22].

3.1. The baseline systems

Two baselines including i-vector [4] and x-vector [10] are evaluated in this study. The results of SITW evaluation are shown in Table 1. The first baseline system i-vector starts from training a Gaussian mixture model (GMM) which is built using a pool of speakers and recordings as the universal background model (UBM). Based on GMM-UBM and an estimated low-rank matrix, i-vector extractor transforms the recording sequence into a fixed dimensional embedding. A 2,048 component gender-independent GMM-UBM with 400-dimensional i-vectors is used as the baseline. In the evaluation, i-vectors are mean centered and followed by LDA and PLDA scoring. The LDA is further applied to decrease the dimensionality to 150 prior to PLDA. The result of baseline i-vector showed EER=5.659%, minDCF1=0.463, and minDCF2=0.629. The second baseline system is a standard x-vector system. The x-vector is based on 7 hidden layers instead of 6 hidden layers of the proposed methods. With the LSTM layer, actually, the training or speaker embedding extraction of the proposed TDNN-LSTM neural network is slower than x-vector. The result of baseline x-vector showed EER=3.499%, minDCF1=0.342, and minDCF2=0.516 in Table 1. The result of x-vector is significantly better than i-vector.

3.2. Effects of different data augmentation methods

Data augmentation is essential to boost the robustness of the speaker or acoustic model, and also to avoid overfitting during the training step. To explore the effects of different data augmentation methods, according to the neural network architecture of TDNN for x-vector, first, we use only the original speech without any data augmentation for training speaker embedding. There are 1,231,170 utterances in the original speech which is the combination of VoxCeleb 1 [9] and VoxCeleb 2 [16]. The results of DCF1=0.428 and DCF2=0.627 are shown in Table 1. Compared with the results of original speech only and x-vector baseline, we found the benefit of data augmentation. By adding various data augmentation methods, we can see the step-by-step improvements including adding room impulse responses (+reverb), adding speed perturbation (+speed), adding volume perturbation (+volume), and adding additive noises (+noise). Finally, with the same neural network architecture of TDNN for x-vectors, the proposed four different data augmentation methods outperform the x-vector baseline which is only two data augmentation methods (+reverb and +noise). The x-vector baseline only considered two data augmentation methods including room impulse responses and additive noises. By using all of the proposed four data augmentation, the DCF1 can be reduced from 0.342 to 0.289 which is the 15.54% relative improvement. The DCF2 can also be reduced from 0.516 to 0.443.

3.3. Effects of TDNN-LSTM neural network embedding

We can further use the proposed TDNN-LSTM neural network embedding with four different data augmentation methods instead of the TDNN based x-vector. We obtain EER=1.859%, DCF1=0.204, and DCF2=0.368 as shown in Table 1. There are two findings. First, by changing the network type from TDNN based x-vector to the proposed TDNN-LSTM, EER is reduced from 3.253% to 1.859%, DCF1 is reduced from 0.289 to 0.204, and DCF2 is reduced from 0.443 to 0.368. Second, compared with the baseline x-vector [10], EER is reduced from 3.499% to 1.859%, DCF1 is reduced from 0.342 to 0.204, and DCF2 is reduced from 0.516 to 0.368. These results denote that the proposed methods outperform the x-vector baseline.

3.4. Computation cost

The proposed data augmentation and TDNN-LSTM neural network embedding methods are trained on NVIDIA TESLA v100 GPUs. After data augmentation, the overall train available data is increased to 16,000 hours. Also, the LSTM itself is more complex than the TDNN neural network. As a result, even using GPUs, it takes around one month to complete the training process. It is worth to open source and trained models. The download available link is at https://sites.google.com/site/chiccoclhuang/sre. Everyone can easily reproduce experimental results in this paper.

4. CONCLUSIONS

The neural network based speaker embedding is a data greedy method. Data augmentation is essential to boost the robustness of the acoustic or speaker model, and also to avoid overfitting during the training step. In this study, we investigated four different data augmentation to increase the amount of available data for centering, LDA, PLDA, and speaker embedding neural networks training. Based on the data augmentation, this paper further explored the TDNN-LSTM neural network embedding speaker representation for improving speaker recognition. The proposed methods were trained on the VoxCeleb dataset including more than 2,000 hours speech and 7,000 speakers. The SITW evaluation set in the core-core condition was evaluated. In total, there were 721,788 trials. Our results showed good improvement on both EER and minimum detection cost function compared with the baseline of i-vector and x-vector. Experimental results confirmed the proposed data augmentation trained TDNN-LSTM speaker embedding representation can provide a robust speaker characterization.

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